UNIST
Department of Physics
Extreme Laser Plasma Photonics & LabAstrophysics

XL-QRH-SPF-001
Revision A
21 September 2026

xLaPPLAs

Spitfire / Empower Raw Commands Quick Reference Handbook

Recovered command catalogue for laboratory operation, software verification and troubleshooting.

STATUS	CONTROLLED INTERNAL REFERENCE
EDITION	Version 1.0
SOFTWARE BASELINE	Spitfire Python Control 1.1.3
LEGACY BASELINE	Spitfire GUI 1.6.0
OWNER	You Jaehyeong
PRINCIPAL INVESTIGATOR	Inhyuk Nam

Independent laboratory reference. Recovered command catalogue.
Not a manufacturer-issued remote programming specification.

xLaPPLAs
Extreme Laser Plasma Photonics & Lab Astrophysics @ UNIST
xlapplas.com

Authority, contents and issue record

FIELD	CONTROLLED VALUE
Document number	XL-QRH-SPF-001
Title	Spitfire / Empower Raw Commands Quick Reference Handbook
Edition	Version 1.0
Revision	A, initial issue
Issue date	21 September 2026
Classification	INTERNAL
Software baseline	Spitfire Python Control 1.1.3
Legacy baseline	Spitfire GUI 1.6.0
Prepared by	You Jaehyeong
Reviewed by	You Jaehyeong
Approved by	You Jaehyeong
Principal Investigator	Inhyuk Nam

Contents

1	REFERENCE CONVENTIONS	3	9	ECU / TEMPERATURE CONTROL	11
2	UDP WIRE FORMAT	4	10	HEAD / COMPRESSOR MOTOR	12
3	CONTROLLER / IDENTIFICATION	5	11	EMPOWER SERVICE PARAMETERS	13
4	EMPOWER OPERATING COMMANDS	6	12	LEGACY / UNRESOLVED ENTRIES	14
5	TDG HIGH VOLTAGE	7	13	RESPONSE INTERPRETATION	15
6	TDG OSCILLATOR / TRIGGER	8	A	QUERY-ID INDEX	16
7	TDG CHANNEL DELAYS	9	B	WRITE-ID INDEX	17
8	TDG STATUS / BWD / OVERRIDE	10	C	SOURCE AND REVISION RECORD	18

STATUS AND AUTHORITY

This handbook is an independent laboratory reference derived from recovered software and legacy records. It does not establish manufacturer specifications, safe operating setpoints, factory limits or hardware compatibility. Unresolved entries remain explicitly marked as unresolved.

Coverage: all 146 catalogue entries—90 IDs in the query range and 56 IDs in the write range. Page numbers are PDF page numbers.

1 REFERENCE CONVENTIONS

1.1 SCOPE

This handbook identifies recovered raw commands, arguments used by the Python implementation, and corresponding readback queries. It covers the ECLIPSE UDP path. It does not define a serial service-port command language.

1.2 READING THE TABLES

TERM	MEANING
ID	Recovered request/reply correlation number. Preserve the listed ID and command spelling. A duplicate command text may have several IDs.
ARG	Text following a write command. NONE means the current caller sends no argument. UNRESOLVED means no validated argument syntax is established.
RB	Readback ID(s) queued by Python after that write. A raw UDP client must issue its own readback queries. RB is not a firmware guarantee of the write effect.
GUI UNIT	A software display label, not independently verified wire scaling. Units without a confirmed conversion must be checked against the controller.
<number>, <ch>	Placeholders; substitute a value and omit the angle brackets. Do not transmit explanatory labels, ARG or RB.

1.3 COMMAND ENTRY

In Programmer, select the ID and enter only the argument. Leave the argument blank for ordinary queries. The application adds the command, ID, UDP header and LF. In a separate UDP client, construct the full packet shown on page 4.

1.4 WRITE AUTHORIZATION AND CONFIRMATION

The application starts DISARMED. Its ARM button permits hardware writes after a recent valid reply. DISARM cancels queued writes; it does not send laser OFF. A raw external client has no application ARM protection. A positive write acknowledgement confirms neither physical output nor completion.

The following IDs are accepted by the application while DISARMED: 2006, 2013, 2016, 2023 and 2028. This is a software exception list, not a declaration that each action is physically safe. For any write, check the relevant readback and the applicable equipment procedure.

1.5 PRF CONVERSION

GUI frequency	Raw argument for ID 2004	Raw query 1010
1,000 Hz	100.00	100.00 maps to 1,000 Hz
Formula	wire value = Hz / 10	display Hz = nonnegative wire value x 10

Programmer uses raw values without this conversion. The factor of ten is established by the recovered GUI and Python conversion code; it is not applied to TDG:TRIG:FREQ.

Source: (S1) commands.py, controller.py, ui.py; (S2) PRF conversion notes.

2 UDP WIRE FORMAT

ITEM	FORMAT
Transport	UDP; one outstanding command at a time in Python.
Legacy remote endpoint	10.19.70.13:1024; configurable in Python.
Legacy local endpoint	0.0.0.0:1024; configurable in Python.
Encoding / limit	ASCII; maximum 1,400 bytes per datagram, including header.
Termination	Each command line ends with LF (hex 0A). No additional NUL byte.
Reply matching	Configured source IP and source port, plus the current request ID.

2.1 COMMAND DATAGRAM

```
Spectra-Physics:CMD_ECLIPSE,[1000]*idn?\n
```

Here and below, \n denotes one LF byte. Do not transmit the two printable characters backslash and n. The comma after CMD_ECLIPSE and the brackets around the numeric ID are required.

2.2 REPLY DATAGRAM

```
Spectra-Physics:REPLY_ECLIPSE:<1000>Spitfire-example\n
```

This identity text is illustrative. Preserve the actual device reply. The reply uses a colon after REPLY_ECLIPSE and angle brackets around the matching ID. Reply lines may be separated by CR/LF.

2.3 REPRESENTATIVE COMMAND LINES

```
[1007]EMP:READ:DIODE:CURR?
[1008]EMP:READ:POW?
[1010]EMP:QSW:PRF?
[2004]EMP:QSW:PRF <raw-number>
```

Each line must be inside an ECLIPSE datagram and terminated by LF. The last line is a syntax template, not an operating instruction. Query and write IDs are distinct even when their base text is similar.

2.4 RS-232 IS A SEPARATE PATH

The raw RS-232 terminal sends the selected ASCII or HEX bytes without ECLIPSE wrapping or command-ID validation. Its COM1 / 19200 / 8-N-1 defaults do not establish the actual service-port protocol. This UDP catalogue must not be assumed valid over RS-232.

Source: (S1) protocol.py, transport.py, serial_service.py; (S2) PROTOCOL.md. (S3) 4-5 / PDF 51 describes Ethernet connectivity, not this packet format.

3 CONTROLLER / IDENTIFICATION

Query aliases are listed explicitly. The reply must retain the ID used in the request.

ID	COMMAND (EXACT SPELLING)	FUNCTION / ARGUMENT / READBACK
1000	`*idn?`	Controller identity. Raw text; also used by the independent connection heartbeat in Python 1.1.3.
1038 1062 1081	`*IDN?`	Controller identity aliases with this exact uppercase spelling.
1082	`*IdN?`	Controller identity spelling variant. Preserve the mixed case.
1016	`EMP:VERSION?`	Empower version query. Distinct recovered spelling from EMP:VERS?.
1034	`HED:VERS?`	Head module firmware/version text.
1035	`EMP:VERS?`	Empower module firmware/version text.
1036	`TDG:VERS?`	Timing/delay generator firmware/version text.
1037	`ECU:VERS?`	ECU/TCU module firmware/version text.
1085	`STATUS?`	Overall controller status. Full bit definition is unresolved.
1086	`MONITOR?`	Monitor-related status. Detailed mode semantics are unresolved.
1089	`COMSTATUS?`	Subsystem communication status. Full bit definition is unresolved.
2027	`ON`	System ON request. Scope must not be assumed identical to EMP:ON. **ARG:** NONE. **RB:** 1013, 1021, 1085.
2028	`OFF`	System OFF request. Python standby separately also sends 2016; no motor STOP is implied. **ARG:** NONE. **RB:** 1013, 1021, 1085.
2050	`RESET`	System RESET request. Exact reset scope is unresolved. **ARG:** NONE in current GUI**RB:** 1089, 1088, 1014, 1066, 1057.
2051	`MONITOR`	Monitor configuration command. Effect and accepted arguments are unresolved. **ARG:** UNRESOLVED. **RB:** 1086.
2055	`CLEAR`	CLEAR command. Exact clear scope is unresolved. **ARG:** UNRESOLVED. **RB:** 1089, 1088, 1014, 1066, 1057.

Source: (S1) commands.py, ui.py, controller.py.

4 EMPOWER OPERATING COMMANDS

Set current and sensed current are separate values. Power readback is not Spitfire compressed-output power.

ID	COMMAND (EXACT SPELLING)	FUNCTION / ARGUMENT / READBACK
1007	`EMP:READ:DIODE:CURR?`	Sensed diode current. GUI unit: A.
1008	`EMP:READ:POW?`	Empower power readback. GUI unit: W; sensing location and physical calibration are not established here.
1009	`EMP:DIODE:CURR?`	Diode-current setpoint. GUI unit: A. Compare sensed current separately using 1007.
1010	`EMP:QSW:PRF?`	Q-switch PRF readback. Nonnegative raw value x 10 gives GUI Hz; see page 3.
1011	`EMP:QSW:PWID?`	Q-switch pulse-width parameter. GUI unit: ns; wire scaling and signal definition remain unverified.
1012	`EMP:QSW:SOURCE?`	Q-switch source code. Physical mode-to-code mapping is unresolved.
1013	`EMP:STB?`	Empower status integer. Partial implementation decoding is described on page 15.
1014	`EMP:FAULT?`	Empower fault record. Python expects four space-separated nonnegative decimal words.
2003	`EMP:DIODE:CURR`	Set diode-current demand. GUI unit: A; operating limits are equipment-specific. **ARG:** <number> **RB:** 1009.
2004	`EMP:QSW:PRF`	Set Q-switch PRF. Use raw frequency value, not GUI Hz. **ARG:** <Hz / 10>, two decimal places **RB:** 1010.
2005	`EMP:ON`	Empower ON request. Read status separately. **ARG:** NONE. **RB:** 1013.
2006	`EMP:OFF`	Empower OFF request. Read status separately. **ARG:** NONE. **RB:** 1013.
2007	`EMP:QSW:PWID`	Set Q-switch pulse-width parameter; definition remains unverified. **ARG:** <number>; GUI label ns **RB:** 1011.
2008	`EMP:QSW:SOURCE`	Set Q-switch source. GUI offers 0/1, but does not establish the physical mode mapping. **ARG:** 0 or 1 in Python GUI **RB:** 1012.
2044	`EMP:RESETFAULTS`	Request Empower fault reset; read the fault record afterward. **ARG:** NONE. **RB:** 1014.

The manufacturer manual describes EXTERNAL, INTERNAL and HOLD OFF. It does not supply a numeric mapping for command 2008; no mapping is inferred here.

Source: (S1) commands.py, ui.py, state.py; (S2) PRF notes; (S3) 6-11 / PDF 71.

5 TDG HIGH VOLTAGE

Enable state, voltage setpoint and measured voltage are different quantities.

ID	COMMAND (EXACT SPELLING)	FUNCTION / ARGUMENT / READBACK
1020	`TDG:VOLT?`	Legacy common TDG voltage query. Channel scope and wire unit are unresolved.
1021 1061	`TDG:HV:ON?`	HV enable status. Parser expects exactly 0 = OFF or 1 = ON; not a residual-voltage measurement.
1040	`TDG:HV1:VOLT?`	HV channel 1 voltage setpoint. GUI unit: V.
1041	`TDG:HV2:VOLT?`	HV channel 2 voltage setpoint. GUI unit: V.
1042	`TDG:READ:HV1VOLT?`	Measured HV channel 1 voltage. GUI unit: V.
1043	`TDG:READ:HV2VOLT?`	Measured HV channel 2 voltage. GUI unit: V.
1055	`TDG:READ:HV1CURR?`	Measured HV channel 1 current. GUI unit: A; physical scaling is unverified.
1056	`TDG:READ:HV2CURR?`	Measured HV channel 2 current. GUI unit: A; physical scaling is unverified.
2015	`TDG:VOLT`	Legacy common TDG voltage setting. Relation to separate channels is unresolved. **ARG:** UNRESOLVED. **RB:** 1020.
2016	`TDG:HV:OFF`	Request HV OFF. Follow enable-state and measured-voltage readbacks separately. **ARG:** NONE. **RB:** 1021.
2017	`TDG:HV:ON`	Request HV ON. An accepted command is not measured voltage confirmation. **ARG:** NONE. **RB:** 1021.
2025	`TDG:HV1:VOLT`	Set HV channel 1 voltage demand. GUI unit: V. **ARG:** <number>**RB:** 1040.
2026	`TDG:HV2:VOLT`	Set HV channel 2 voltage demand. GUI unit: V. **ARG:** <number>**RB:** 1041.

CONFIRMATION REFERENCE

After 2016 or 2017, Python automatically queries 1021. For voltage readback, query 1042 and 1043 explicitly, or use enabled HV monitoring. A reply to 2016 itself must not be substituted for a reply to 1021.

Source: (S1) commands.py:readback_ids, state.py, ui.py.

6 TDG OSCILLATOR / TRIGGER

Oscillator synchronization, amplifier triggering and Empower Q-switch source are separate controls.

ID	COMMAND (EXACT SPELLING)	FUNCTION / ARGUMENT / READBACK
1024	`TDG:OSC:FREQ?`	Oscillator frequency setting/reference. Distinct from measured frequency 1065; wire unit unverified.
1025	`TDG:DIV?`	Repetition-rate divisor. Read the returned value; valid range is configuration-specific.
1050 1067	`TDG:OSC:MODE?`	Oscillator synchronization mode. Parser accepts INT/EXT or numeric responses; hardware encoding requires confirmation.
1051	`TDG:TRIG:MODE?`	Amplifier trigger mode. Same software text choices; not the Q-switch source enum.
1053	`TDG:TRIG:FREQ?`	Internal amplifier trigger frequency. GUI unit: Hz; no PRF factor-of-ten conversion.
1065	`TDG:READ:OSCQUEST?`	Measured oscillator frequency. GUI unit: MHz.
1079	`TDG:TRIG?`	Separate TDG trigger-related query. Detailed response semantics are unresolved.
1080	`TDG:OSC:SETMODE?`	TDG operation mode. GUI choices: CONT, COUNT, COUNT 1. Count/gating behavior is unverified.
1083	`TDG:DELMON?`	Delay-monitor-related query. Response definition is unresolved.
2020	`TDG:DIV`	Set repetition-rate divisor. Manual describes integer division; validate the required integer range. **ARG:** <integer>**RB:** 1025.
2030	`TDG:OSC:MODE`	Set oscillator synchronization source. **ARG:** INT or EXT as used by Python.**RB:** 1050.
2031	`TDG:TRIG:MODE`	Set amplifier trigger source. **ARG:** INT or EXT as used by Python.**RB:** 1051.
2032	`TDG:TRIG:FREQ`	Set internal amplifier trigger frequency. GUI unit: Hz. **ARG:** <number>**RB:** 1053.
2046	`TDG:OSC:SETTOFRQ`	Apply measured oscillator value to the setting/reference. Current GUI supplies the numeric value from 1065. **ARG:** <number>; not argument-free **RB:** 1024, 1065.
2048	`TDG:OSC:SETMODE`	Set TDG operation mode. Keep the space in COUNT 1. **ARG:** CONT, COUNT, or COUNT 1.**RB:** 1080.

Source: (S1) ui.py:_oscillator_to_measured, commands.py, state.py; (S3) 6-12 to 6-13 / PDF 72-73.

7 TDG CHANNEL DELAYS

These are logical channel identifiers. Do not infer a physical connector mapping for every channel.

ID	COMMAND (EXACT SPELLING)	FUNCTION / ARGUMENT / READBACK
1045 1058 1063	`TDG:DEL1?`	Logical delay channel 1. Python decodes numeric value and separate mode flags.
1046 1059 1064	`TDG:DEL2?`	Logical delay channel 2. Python decodes numeric value and separate mode flags.
1047	`TDG:DEL3?`	Logical delay channel 3.
1048	`TDG:DEL4?`	Logical delay channel 4.
1049	`TDG:DEL5?`	Logical delay channel 5. Physical output mapping is unresolved.
1052	`TDG:DEL0?`	Logical delay 0. Separate from the GUI channel 1-5 controls.
2019	`TDG:DEL`	Set one channel delay value or one mode token. Validated firmware ranges are not provided. **ARG:** <ch> <value-or-token>; ch = 1.**RB:** 1045-1049 by channel.
2054	`RES12`	RES12 command. Follow-up queries target delay 1 and 2; exact reset effect is unresolved. **ARG:** UNRESOLVED. **RB:** 1045, 1046.

7.1 ARGUMENT FORMS USED BY PYTHON

ARGUMENT FORM	INTERPRETATION
<ch> <number>	Numeric delay; wire units and range must be established for the target controller.
<ch> on / <ch> off	Switch token; Python sends lowercase on/off.
<ch> BWD / <ch> NBWD	BWD-related mode token. Detailed protection behavior is unresolved.
<ch> ENA / <ch> NENA	Enable-related mode token. Detailed gating behavior is unresolved.

7.2 COMPOUND REPLY EXAMPLE

```
<1045>ON NBWD NENA DELAY VALUE = 12.50
```

Synthetic parser fixture, not a captured hardware reply. Python separates ON/OFF, BWD/NBWD and ENA/NENA, and reads the number following six prefix fields. A bare number does not establish any mode flags.

Source: (S1) ui.py:_apply_delay / _apply_channel_mode, state.py:decode_delay; (S2) legacy delay-parser notes.

8 TDG STATUS / BWD / OVERRIDE

ID	COMMAND (EXACT SPELLING)	FUNCTION / ARGUMENT / READBACK
1019	`TDG:STB?`	TDG status integer. Communication-error mask used by Python: 0x800.
1022	`TDG:STAT:BWD?`	BWD status query. STAT:BWD payload definition is unresolved.
1023 1039	`TDG:STBH?`	TDG extended status query aliases. Full bit definition is unresolved.
1027 1054	`TDG:BWD?`	BWD-related readback. Detailed enum or bit definition is unresolved.
1066	`TDG:STBHFLT?`	TDG extended fault/status readback. Preserve the raw returned value.
1087	`TDG:FAULTOVERRIDE?`	TDG fault-override flag. Python expects 0/1; firmware effect requires verification.
2018	`TDG:RESBWD`	Request BWD reset. A reset acknowledgement is not evidence that the underlying condition is resolved. **ARG:** NONE. **RB:** 1054, 1066.
2033	`TDG:BWD`	Set BWD-related configuration. Accepted arguments and effect are unresolved. **ARG:** UNRESOLVED. **RB:** 1054.
2052	`TDG:RESFAULT`	Request TDG fault reset. **ARG:** NONE. **RB:** 1066.
2053	`TDG:FAULTOVERRIDE`	Set TDG fault override. Changes a protection-related configuration. **ARG:** 0 or 1 in Python GUI**RB:** 1087.

BWD REFERENCE

The manufacturer describes the bandwidth detector as monitoring the stretcher spectrum. Its reset control restores the relay after the underlying condition is corrected. That description does not define all bits or tokens in the recovered BWD command family.

RAW STATUS RETENTION

Keep the complete response and its ID when investigating a fault. A zero value in one status field must not be treated as proof that every interlock is satisfied. The partial status decoding implemented by Python is on page 15.

Source: (S1) commands.py, state.py; (S3) 6-10 / PDF 70.

9 ECU / TEMPERATURE CONTROL

ECU command names correspond to the GUI temperature-control module (TCU).

ID	COMMAND (EXACT SPELLING)	FUNCTION / ARGUMENT / READBACK
1028	`ECU:STB?`	ECU/TCU status integer. Communication-error mask used by Python: 0x800.
1029 1057 1088	`ECU:STBH?`	ECU extended/health-status aliases. Full bit definition is unresolved.
1030	`ECU:READTEMP?`	Crystal-temperature readback. GUI unit: degrees C; sensor location is unverified.
1031	`ECU:HUM?`	Humidity readback. GUI unit: %RH; sensor location and range are unverified.
1032	`ECU:TEMP?`	Temperature setpoint. GUI unit: degrees C; distinct from the measured temperatures.
1033	`ECUOPTOUT?`	ECU opt-out flag. Python GUI uses 0/1; participation/protection effect requires verification.
1044	`ECU:HUMOVERRIDE?`	Humidity-override flag. Python GUI uses 0/1.
1084	`ECU:TECTEMP?`	TEC-temperature readback. GUI unit: degrees C; distinct from Empower LBO temperature.
2021	`ECU:TEMP`	Set ECU temperature demand. GUI unit: degrees C. **ARG:** <number>**RB:** 1032.
2022	`ECU:TECON`	Request TEC ON. Not a command to start an external chiller. **ARG:** NONE. **RB:** 1028.
2023	`ECU:TECOFF`	Request TEC OFF. This may interrupt temperature control. **ARG:** NONE. **RB:** 1028.
2024	`ECUOPTOUT`	Set ECU opt-out. Protection/module participation setting. **ARG:** 0 or 1 in Python GUI**RB:** 1033.
2029	`ECU:HUMOVERRIDE`	Set humidity override. Protection-related setting. **ARG:** 0 or 1 in Python GUI**RB:** 1044.
2049	`ECU:RESETFAULTS`	Request ECU fault reset. **ARG:** NONE. **RB:** 1057.

Source: (S1) commands.py, state.py, ui.py. GUI units are recorded as labels, not independent calibrations.

10 HEAD / COMPRESSOR MOTOR

ID	COMMAND (EXACT SPELLING)	FUNCTION / ARGUMENT / READBACK
1015	`HED:STB?`	Head/compressor status integer. Communication-error mask used by Python: 0x800.
1017	`HED:MOTOR:*IDN?`	Head motor identity query. Preserve the raw returned text.
1018 1060	`HED:MOTOR:SPEED?`	Motor speed query aliases. GUI label: %. Correspondence to the legacy speed modes is unresolved.
1069	`HEDOPTOUT?`	Head/compressor opt-out flag. Python GUI uses 0/1.
2009	`HED:MOTOR:REW`	Request reverse motion. Direction and continuation behavior are equipment-specific. **ARG:** NONE. **RB:** 1015.
2010	`HED:MOTOR:FWD`	Request forward motion. Direction and continuation behavior are equipment-specific. **ARG:** NONE. **RB:** 1015.
2011	`HED:MOTOR:JOGF`	Request a forward jog. Distance is not established by the catalogue. **ARG:** NONE. **RB:** 1015.
2012	`HED:MOTOR:JOGR`	Request a reverse jog. Distance is not established by the catalogue. **ARG:** NONE. **RB:** 1015.
2013	`HED:MOTOR:STOP`	Request motor STOP. Separate from the system OFF and HV OFF commands. **ARG:** NONE. **RB:** 1015.
2014	`HED:MOTOR:SPEED`	Set motor speed parameter. GUI label: %; mapping to actual speed is unresolved. **ARG:** <number>**RB:** 1060.
2035	`HEDOPTOUT`	Set head/compressor opt-out. Protection/module participation setting. **ARG:** 0 or 1 in Python GUI**RB:** 1069.

MOTION COMMAND LIFETIME

The legacy manual describes continuous REV/FWD motion while a mouse button is held. The Python motor buttons send one command per click and do not automatically send STOP on mouse release. A raw command likewise has no mouse-release event. Use the actual controller behavior and motion procedure to define the stop action.

The manual describes four speed choices. The Python percentage label does not establish a linear percentage-to-speed conversion.

Source: (S1) ui.py, commands.py; (S3) 6-9 to 6-10 / PDF 69-70.

11 EMPOWER SERVICE PARAMETERS

Recovered labels do not establish firmware limits, calibration equations or persistence.

QUERY / WRITE	COMMAND	SERVICE ITEM / ARGUMENT
1070 Q 2037 W	EMP:SECRETMSG1? EMP:SECRETMSG1	**Unassigned service item** GUI label: unresolved. Function and accepted argument are unresolved. **ARG:** UNRESOLVED. **RB:** 1070.
1071 Q 2038 W	EMP:SECRETMSG2? EMP:SECRETMSG2	**Max current** GUI label: A. Service label only; equipment-specific limit and storage behavior require confirmation. **ARG:** <number>**RB:** 1071.
1072 Q 2039 W	EMP:SECRETMSG3? EMP:SECRETMSG3	**Max power** GUI label: mW. Actual scale and role are unresolved: limit, fault threshold or calibration cannot be distinguished from this label. **ARG:** <number>**RB:** 1072.
1073 Q 2040 W	EMP:SECRETMSG4? EMP:SECRETMSG4	**Intercept** GUI label: mW. Calibration equation, signal source and range are unresolved. **ARG:** <number>**RB:** 1073.
1074 Q 2041 W	EMP:SECRETMSG5? EMP:SECRETMSG5	**Slope**. GUI label: mW/V. Calibration equation, signal source and range are unresolved. **ARG:** <number>**RB:** 1074.
1075 Q 2042 W	EMP:SECRETMSG6? EMP:SECRETMSG6	**Over-temperature limit** GUI label: degrees C. Monitored object and physical threshold behavior are unresolved. **ARG:** <number>**RB:** 1075.
1076 Q 2043 W	EMP:SECRETMSG7? EMP:SECRETMSG7	**Under-temperature limit** GUI label: degrees C. Monitored object and physical threshold behavior are unresolved. **ARG:** <number>**RB:** 1076.

ID	COMMAND (EXACT SPELLING)	FUNCTION / ARGUMENT / READBACK
1077	EMP:LBOTEMP?	LBO temperature readback. GUI unit: degrees C.
1078	EMP:LBOBREAK?	LBO stability fault-break readback; firmware polarity/effect unverified.
2047	EMP:LBOBREAK	Set LBO stability fault-break parameter. **ARG:** 0 or 1 in Python GUI**RB:** 1078.
1068	EMPOPTOUT?	Empower opt-out flag. Python GUI uses 0/1.
2036	EMPOPTOUT	Set Empower opt-out. Protection/module participation setting. **ARG:** 0 or 1 in Python GUI**RB:** 1068.
2045	EMP:SECRETSCOMT	Service commit request. Nonvolatile retention and power-cycle behavior are unverified. **ARG:** NONE. **RB:** not assigned.

Programmer does not automatically append commit 2045 to a service write. The dedicated service-value GUI does. A raw readback of 6 for Max power establishes only a stored parameter value; it does not establish a 6 mW or 6 W output ceiling.

Source: (S1) ui.py:_apply_service_value, commands.py. Units above are GUI labels.

12 LEGACY / UNRESOLVED ENTRIES

Listed for catalogue completeness. Unknown syntax or effect is not replaced with a guessed procedure.

ID	COMMAND (EXACT SPELLING)	FUNCTION / ARGUMENT / READBACK
1001	`MIL:DIODE1:CURR?`	MIL diode 1 current-setting query by name. Target module and unit are unverified.
1002	`MIL:DIODE2:CURR?`	MIL diode 2 current-setting query by name. Target module and unit are unverified.
1003	`MIL:POW?`	MIL power-setting query by name. Target module and unit are unverified.
1004	`MIL:READ:DIODE1:CURR?`	MIL diode 1 current-readback query by name. Physical quantity and unit are unverified.
1005	`MIL:READ:DIODE2:CURR?`	MIL diode 2 current-readback query by name. Physical quantity and unit are unverified.
1006	`MIL:READ:POW?`	MIL power-readback query by name. Do not identify it as Spitfire final output.
2000	`MIL:DIODE1:CURR`	MIL diode 1 current-setting family. Function and argument specification are unverified. **ARG:** UNRESOLVED. **RB:** 1001.
2001	`MIL:DIODE2:CURR`	MIL diode 2 current-setting family. Function and argument specification are unverified. **ARG:** UNRESOLVED. **RB:** 1002.
2002	`MIL:READ:POW`	Classified as a write by ID despite READ in the command text. Effect is unresolved. **ARG:** UNRESOLVED. **RB:** not assigned.
1026	`TDG:DEL`	HOLD: no question mark; classified in the query range only by the recovered ID. Read-only behavior is not established.
2034	`[EMPTY STRING]`	HOLD: the recovered command string is empty. Do not construct a command by guessing missing text. **ARG:** UNRESOLVED; do not use. **RB:** not assigned.

CLASSIFICATION LIMIT

IDs 1000-1089 occupy the query range; IDs 2000-2055 occupy the write range. Numeric range and SCPI-like spelling are software catalogue conventions, not a substitute for a validated firmware specification. In particular, ID 1026 is not established as a harmless query.

Source: (S1) commands.py; (S2) recovered catalogue. MIL support for the installed system is not established.

13 RESPONSE INTERPRETATION

13.1 ACCEPTANCE AND READBACK

Python accepts exact OK or ACK as positive write acknowledgement text. ERR/ERROR, NACK and FAIL/FAILED prefixes are errors; other write acknowledgements are not accepted as success. An error or malformed numeric reply must not be substituted with zero.

A timeout means that a matching reply was not received in time. It does not prove that a preceding write had no effect. Preserve the request, raw response and timestamps before deciding whether a write should be repeated.

13.2 MODULE COMMUNICATION STATUS

QUERY	MODULE
1013 / EMP:STB?	Empower
1015 / HED:STB?	Head / compressor
1019 / TDG:STB?	Timing / delay generator
1028 / ECU:STB?	ECU / temperature control

For these four integer status words, Python uses status & 0x800 as the communication-error test. This is a partial legacy-derived mapping. A clear bit does not establish that all other status conditions are normal. Fresh module replies are required; a fresh controller identity alone is insufficient.

13.3 EMP:FAULT? FOUR-WORD RECORD

```
<1014>w0 w1 w2 w3
```

Python expects four space-separated nonnegative decimal integers. w0 is the first returned word. The following is the partial implemented indicator map; it is not a complete firmware fault dictionary.

WORD	MASK	IMPLEMENTED INDICATOR
w0	0x08	Empower: driver overtemperature
w0	0x10	Empower: underfrequency
w0	0x20	Empower: overcurrent
w0	0x40	TDG: user interlock
w1	0x02	Empower: VSWR
w1	0x04	Empower: keyswitch
w1	0x08	Empower: checksum
w1	0x10	Empower: array open
w1	0x20	TDG: head interlock
w1	0x40	TCU: TEC overtemperature
w1	0x80	Empower: cover
w2	0x02	TCU communication
w2	0x10	Empower: protection
w3	0xFF	Empower: LBO under-temperature

A set mask is reported as active. For w3 / 0xFF, the implementation tests the low byte for a nonzero LBO under-temperature code. Unlisted bits and unresolved indicators remain undefined.

Source: (S1) state.py:apply, is_error_reply, communication_state, FAULT_WORD_BITS.

A QUERY-ID INDEX

Exact catalogue spelling. Page references point to the descriptive command table.

ID	COMMAND	PAGE
1000	*idn?	5
1001	MIL:DIODE1:CURR?	14
1002	MIL:DIODE2:CURR?	14
1003	MIL:POW?	14
1004	MIL:READ:DIODE1:CURR?	14
1005	MIL:READ:DIODE2:CURR?	14
1006	MIL:READ:POW?	14
1007	EMP:READ:DIODE:CURR?	6
1008	EMP:READ:POW?	6
1009	EMP:DIODE:CURR?	6
1010	EMP:QSW:PRF?	6
1011	EMP:QSW:PWID?	6
1012	EMP:QSW:SOURCE?	6
1013	EMP:STB?	6
1014	EMP:FAULT?	6
1015	HED:STB?	12
1016	EMP:VERSION?	5
1017	HED:MOTOR:*IDN?	12
1018	HED:MOTOR:SPEED?	12
1019	TDG:STB?	10
1020	TDG:VOLT?	7
1021	TDG:HV:ON?	7
1022	TDG:STAT:BWD?	10
1023	TDG:STBH?	10
1024	TDG:OSC:FREQ?	8
1025	TDG:DIV?	8
1026	TDG:DEL	14
1027	TDG:BWD?	10
1028	ECU:STB?	11
1029	ECU:STBH?	11
1030	ECU:READTEMP?	11
1031	ECU:HUM?	11
1032	ECU:TEMP?	11
1033	ECUOPTOUT?	11
1034	HED:VERS?	5
1035	EMP:VERS?	5
1036	TDG:VERS?	5
1037	ECU:VERS?	5
1038	*IDN?	5
1039	TDG:STBH?	10
1040	TDG:HV1:VOLT?	7
1041	TDG:HV2:VOLT?	7
1042	TDG:READ:HV1VOLT?	7
1043	TDG:READ:HV2VOLT?	7
1044	ECU:HUMOVERRIDE?	11
1045	TDG:DEL1?	9
1046	TDG:DEL2?	9
1047	TDG:DEL3?	9
1048	TDG:DEL4?	9
1049	TDG:DEL5?	9
1050	TDG:OSC:MODE?	8
1051	TDG:TRIG:MODE?	8
1052	TDG:DEL0?	9
1053	TDG:TRIG:FREQ?	8
1054	TDG:BWD?	10
1055	TDG:READ:HV1CURR?	7
1056	TDG:READ:HV2CURR?	7
1057	ECU:STBH?	11
1058	TDG:DEL1?	9
1059	TDG:DEL2?	9
1060	HED:MOTOR:SPEED?	12
1061	TDG:HV:ON?	7
1062	*IDN?	5
1063	TDG:DEL1?	9
1064	TDG:DEL2?	9
1065	TDG:READ:OSCQUEST?	8
1066	TDG:STBHFLT?	10
1067	TDG:OSC:MODE?	8
1068	EMPOPTOUT?	13
1069	HEDOPTOUT?	12
1070	EMP:SECRETMSG1?	13
1071	EMP:SECRETMSG2?	13
1072	EMP:SECRETMSG3?	13
1073	EMP:SECRETMSG4?	13
1074	EMP:SECRETMSG5?	13
1075	EMP:SECRETMSG6?	13
1076	EMP:SECRETMSG7?	13
1077	EMP:LBOTEMP?	13
1078	EMP:LBOBREAK?	13
1079	TDG:TRIG?	8
1080	TDG:OSC:SETMODE?	8
1081	*IDN?	5
1082	*IdN?	5
1083	TDG:DELMON?	8
1084	ECU:TECTEMP?	11
1085	STATUS?	5
1086	MONITOR?	5
1087	TDG:FAULTOVERRIDE?	10
1088	ECU:STBH?	11
1089	COMSTATUS?	5

ID 1026 has no question mark and remains unresolved; its position in this index does not establish read-only behavior.

B WRITE-ID INDEX

Exact catalogue spelling. Page references point to the descriptive command table.

ID	COMMAND	PAGE
2000	MIL:DIODE1:CURR	14
2001	MIL:DIODE2:CURR	14
2002	MIL:READ:POW	14
2003	EMP:DIODE:CURR	6
2004	EMP:QSW:PRF	6
2005	EMP:ON	6
2006	EMP:OFF	6
2007	EMP:QSW:PWID	6
2008	EMP:QSW:SOURCE	6
2009	HED:MOTOR:REW	12
2010	HED:MOTOR:FWD	12
2011	HED:MOTOR:JOGF	12
2012	HED:MOTOR:JOGR	12
2013	HED:MOTOR:STOP	12
2014	HED:MOTOR:SPEED	12
2015	TDG:VOLT	7
2016	TDG:HV:OFF	7
2017	TDG:HV:ON	7
2018	TDG:RESBWD	10
2019	TDG:DEL	9
2020	TDG:DIV	8
2021	ECU:TEMP	11
2022	ECU:TECON	11
2023	ECU:TECOFF	11
2024	ECUOPTOUT	11
2025	TDG:HV1:VOLT	7
2026	TDG:HV2:VOLT	7
2027	ON	5
2028	OFF	5
2029	ECU:HUMOVERRIDE	11
2030	TDG:OSC:MODE	8
2031	TDG:TRIG:MODE	8
2032	TDG:TRIG:FREQ	8
2033	TDG:BWD	10
2034	[EMPTY STRING]	14
2035	HEDOPTOUT	12
2036	EMPOPTOUT	13
2037	EMP:SECRETMSG1	13
2038	EMP:SECRETMSG2	13
2039	EMP:SECRETMSG3	13
2040	EMP:SECRETMSG4	13
2041	EMP:SECRETMSG5	13
2042	EMP:SECRETMSG6	13
2043	EMP:SECRETMSG7	13
2044	EMP:RESETFAULTS	6
2045	EMP:SECRETSCOMT	13
2046	TDG:OSC:SETTOFRQ	8
2047	EMP:LBOBREAK	13
2048	TDG:OSC:SETMODE	8
2049	ECU:RESETFAULTS	11
2050	RESET	5
2051	MONITOR	5
2052	TDG:RESFAULT	10
2053	TDG:FAULTOVERRIDE	10
2054	RES12	9
2055	CLEAR	5

INDEX NOTES

All 56 write-range entries are retained, including the empty string at 2034. Use the descriptive table for arguments and readback IDs. NONE and UNRESOLVED in those tables have different meanings.

The index lists base command text only. A write may require an argument after one space. The ID and command must then be wrapped in the complete ECLIPSE packet; see page 4.

C SOURCE AND REVISION RECORD

C.1 SOURCE HIERARCHY

(S1) Spitfire Python Control 1.1.3. Supplied distribution: Spitfire_Python_Control_1.1.3.zip. Exact command strings and post-write readback mapping: commands.py. Arguments and display labels: ui.py. Packet format: protocol.py and transport.py. Reply interpretation: state.py. Authorization: controller.py. These files define the implementation described here.

(S2) Legacy GUI 1.6.0 recovery recordPROTOCOL.md, REVERSE_ENGINEERING_REPORT.md and CHANGELOG_1_1_1.md within (S1). These record static recovery from SPITFIRE.EXE / SPITFIRE.UIR. This edition uses the recorded analysis; the original executable was not re-executed to create this handbook.

(S3) Spectra-Physics Spitfire Pro User's Manual, 330A, Rev. A. The supplied 110-page PDF provides equipment and operator context. It does not provide the complete recovered numeric-ID command catalogue or the ECLIPSE UDP packet specification.

MANUAL TOPIC	PRINTED PAGE	PDF PAGE
Ethernet connection	4-5	51
Compressor motor / BWD	6-9 to 6-10	69-70
Empower source / current	6-11	71
Timing / divisor / HV	6-12 to 6-13	72-73
Diode and driver faults	6-14 to 6-15	74-75

(S4) Spitfire Python Control 1.1.2 Korean Manual. Used as a secondary editorial reference. Current 1.1.3 source takes precedence where an earlier description differs.

C.2 INTERPRETATION BOUNDARY

Command recovery establishes strings and software behavior. It does not establish the installed firmware's accepted ranges, calibration, physical signal paths, protection behavior or nonvolatile storage rules. Service labels such as Max power must not be promoted to verified physical output limits.

The manufacturer distinguishes diode-pump overpower and driver-FET overpower. Neither is mapped here to EMP:SECRETMSG3, because the available sources do not establish that relationship.

C.3 REVISION RECORD

VERSION / REV	DATE	DESCRIPTION
1.0 / A	21 Sep 2026	Initial English command-only handbook. 146 entries indexed; GUI/wire units distinguished. Corrected the 2046 argument description against the current caller; delay aliases documented against the current parser.

C.4 BASELINE IDENTIFICATION

SHA-256 of the software archive used for this edition:

`bf66ac1d2e14b4b52d901dd946a9dffc`
`709bd5e430125c2a5dde7e7c6e782931`